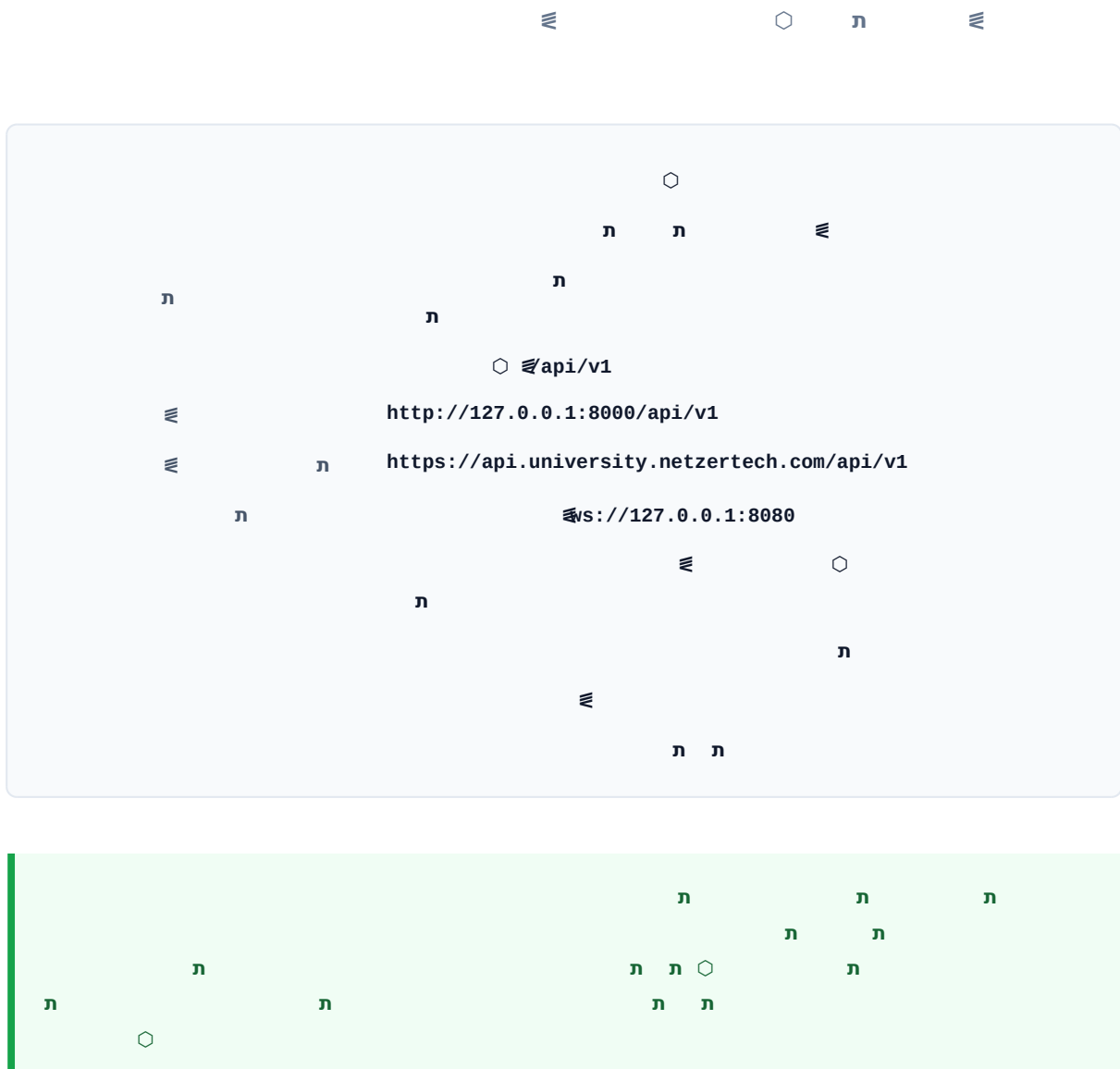
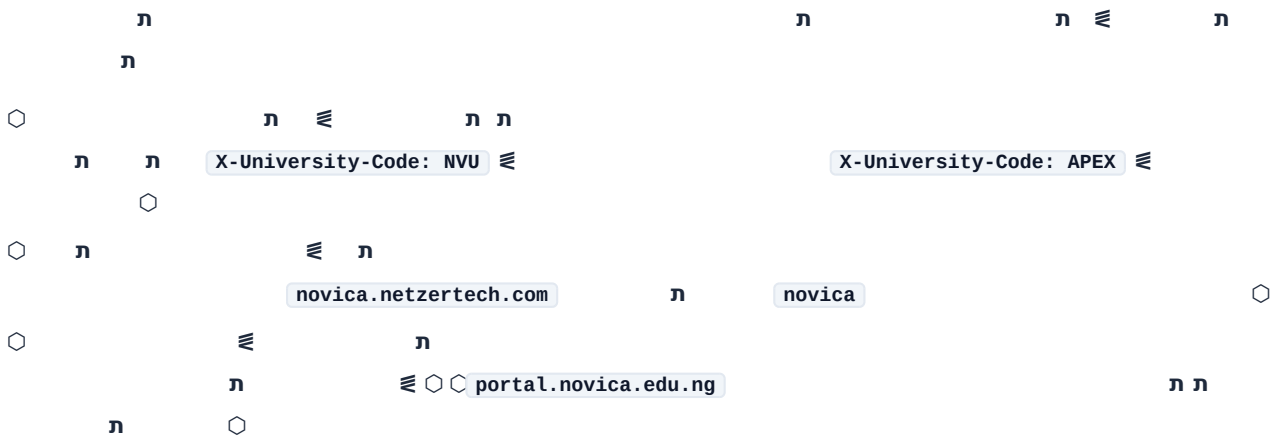


Frontend Engineering API Integration Specification & Step-by-Step Implementation Blueprint





```
// Dynamic CSS Theme Injection in React / Vue
function applyTenantTheme(university) {
  document.documentElement.style.setProperty('--color-primary', university.primary_color || '#2E5FA3');
  document.documentElement.style.setProperty('--color-secondary', university.secondary_color ||
  '#1A7A6E');
  document.title = `${university.name} - Portal`;
}
```

NetzerTech



NetzerTech



Service	URL	Environment Variables
netzertech_nginx	http://localhost:8000	
netzertech_api	localhost:9000	
netzertech_db	localhost:5432	secret_password postgres netzertech_university
netzertech_redis	localhost:6379	
netzertech_reverb	ws://localhost:8080	
netzertech_horizon		



```
git clone https://github.com/NetzerTech-Worldwide/Netzertech-University-Backend.git
cd Netzertech-University-Backend
```



```
docker compose up -d --build
```



```
docker compose exec app php artisan migrate:fresh --seed
```



```
http://localhost:8000/api/v1/courses
```

```
X-University-Code: NVU
```

NetzerTech

```
# View real-time aggregated logs of all services:
docker compose logs -f

# Run automated test suite inside container (61 tests, 406 assertions):
docker compose exec app php artisan test

# Re-seed demo database at any time:
docker compose exec app php artisan migrate:fresh --seed
```

```
# Stop all containers:
docker compose down
```



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```
// Success Envelope (HTTP 200 / 201)
{
  "success": true,
  "message": "Human-readable status summary",
  "data": { ... } // Object or Array
}

// Error Envelope (HTTP 400 / 401 / 403 / 422 / 500)
{
  "success": false,
  "message": "Specific error description",
  "errors": {
    "field_name": ["Validation error message 1", "Validation error message 2"]
  }
}
```



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```

import axios from 'axios';

const apiClient = axios.create({
  baseURL: process.env.VITE_API_BASE_URL || 'http://127.0.0.1:8000/api/v1',
  headers: {
    'Accept': 'application/json',
    'Content-Type': 'application/json',
  },
  timeout: 30000,
});

// Request Interceptor: Injects Sanctum Bearer Token and Tenant Header
apiClient.interceptors.request.use((config) => {
  const token = localStorage.getItem('auth_token');
  const universityCode = localStorage.getItem('tenant_code') || 'NVU';

  if (token) {
    config.headers.Authorization = `Bearer ${token}`;
  }
  config.headers['X-University-Code'] = universityCode;
  return config;
});

// Response Interceptor: Centralized Session Expiry & Error Handling
apiClient.interceptors.response.use(
  (response) => response.data,
  (error) => {
    if (error.response?.status === 401) {
      localStorage.removeItem('auth_token');
      localStorage.removeItem('user_data');
      window.location.href = '/login?expired=1';
    }
    return Promise.reject(error.response?.data || error);
  }
);

export default apiClient;

```



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	/auth/login	{ identifier, password }	≡ ת
	/auth/logout		ת ≡
	/auth/me		ת ≡
	/auth/change-password	{ current_password, new_password, new_password_confirmation }	ת ≡

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○ ת POST /auth/login ת { identifier, password } ○

○ data.token ת data.user.roles ○

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- student → ת /student/dashboard
- postgraduate → ת /postgraduate/dashboard
- hod dean → ת /staff/academics
- applicant → ת /admissions/status
- super_admin → ת /admin/tenants

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	/admissions/programmes		n
	/admissions/register	{ name, email, phone, password }	
	/admissions/steps/bio	{ dob, gender, state, lga, address }	n
	/admissions/steps/academic	{ jamb_reg_no, jamb_score, olevel_results }	n
	/admissions/post-utme/book	{ slot_id }	
	/admissions/submit		
	/admissions/status		
	/admissions/matriculate		

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POST /admissions/submit



```
GET /admissions/status
```

```
admission_status: "admitted"
```

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POST /admissions/matriculate

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	/courses	level, semester	ת
	/courses/register	{ session, semester, course_ids: [] }	ת – ת
	/courses/my-registration	session, semester	ת
	/courses/registrations/{id}/approve	{ remarks }	ת ≡

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	/courses/{id}/materials		ת
	/courses/{id}/materials	{ title, file_url, file_type, size_bytes }	
	/courses/{id}/assignments		ת
	/assignments/{id}/submit	{ submission_text, file_url }	ת
	/assignments/submissions/{id}/grade	{ score, feedback }	

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	/cbt/exams		т т
	/cbt/exams/{id}/start		т т
	/cbt/sessions/{id}/heartbeat	{ current_question_id, selected_option, seconds_remaining }	т
	/cbt/sessions/{id}/submit	{ answers: { "question_id": "option" } }	т

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т document.documentElement.requestFullscreen() т

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POST /cbt/sessions/{id}/heartbeat

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visibilitychange

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	/results/upload	{ student_id, course_id, session, semester, ca_score, exam_score }	т ≡
	/results/semester	session, semester	т
	/results/transcript		т т

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```

async function downloadTranscript() {
  const response = await axios.get('/api/v1/results/transcript', {
    responseType: 'blob',
    headers: { Authorization: `Bearer ${token}` }
  });
  const fileUrl = window.URL.createObjectURL(new Blob([response.data], { type: 'application/pdf' }));
  const link = document.createElement('a');
  link.href = fileUrl;
  link.setAttribute('download', 'Official_Academic_Transcript.pdf');
  document.body.appendChild(link);
  link.click();
  link.remove();
}

```

```

n
  n      <script src="https://js.paystack.co/v1/inline.js"></script>
  POST /finance/pay      n      { access_code, reference, public_key, amount_kobo,
subaccount }
  n      n      callback(response)      POST /finance/verify/{reference}
  n      n

```

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⌵ lockForUpdate()

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	/facilities/hostels		т
	/facilities/hostels/{id}/rooms		т
	/facilities/hostels/reserve-bed	{ bed_space_id }	
	/facilities/hostels/my-allocation		т
	/facilities/hostels/agreement	{ allocation_id, agreed: true }	т т
	/facilities/hostels/maintenance	{ hostel_id, category, description }	

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	/facilities/clinic/register	{ blood_group, genotype, allergies, emergency_contact_name, emergency_contact_phone }	т
	/facilities/clinic/card		т т
	/facilities/clinic/appointments	{ reason, appointment_date }	т
	/facilities/clinic/appointments		т т т

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	/facilities/library/catalog	query, category	ת ת ת
	/facilities/library/reserve	{ library_item_id }	ת
	/facilities/library/my-borrowed		ת ת ת
	/facilities/library/loans/{id}/return		

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	/approvals/requests	{ type, title, reason }	
	/approvals/requests	filter=pending_my_action	
	/approvals/requests/{id}		
	/approvals/requests/{id}/action	{ action: "approved" "rejected", comments }	ת

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	/identity/card		≡ ת
	/identity/card/pdf		≡ ת
	/verify/id-card/{hash}		≡ ת

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`https://portal.university.edu.ng/verify/id-card/{token_hash}` 

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☐	/postgraduate/dashboard		
☐	/postgraduate/proposals	{ title, abstract, document_url }	ת
☐	/postgraduate/proposals/{id}/review	{ status, reviewer_feedback, defense_score, defense_date }	ת
☐	/postgraduate/milestones		–
☐	/postgraduate/milestones/{id}/submit	{ submission_url }	ת ת ת
☐	/postgraduate/milestones/{id}/review	{ status, supervisor_comments }	ת
☐	/postgraduate/supervision-logs	{ staff_id, meeting_date, summary_notes, next_deliverables }	
☐	/postgraduate/supervision-logs/{id}/confirm		
☐	/postgraduate/early-warning		ת ת

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• ת risk_level: "low" ☰ "medium" ☰ "high" ☰ ת ☐

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API

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API	API	API	API
	/ai/advisor/chat	{ prompt, session_id?: string }	API
	/ai/advisor/history	session_id?: string	
	/ai/advisor/clear	{ session_id?: string }	

API

API

POST /ai/advisor/chat

```
{
  "session_id": "session_a8b29c...",
  "response": "Hello Chidi, your current CGPA is 4.33 (First Class Honours)...",
  "suggested_actions": [
    "Review CSC 301 prerequisite requirements",
    "Calculate minimum grade needed in CSC 305",
    "View university academic calendar"
  ],
  "telemetry": { "cgpa": 4.33, "standing": "First Class Honours", "registered_credits": 18 }
}
```

API

suggested_actions



API

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	/career/profile		
	/career/skills	{ name, category, proficiency_level }	תת
	/career/projects	{ title, description, technologies, github_url, live_url }	תת
	/career/certifications	{ title, issuer, issue_date, credential_url }	תת ת ת
	/career/jobs	job_type?: string	ת ת
	/career/cv/pdf		ת ת

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	/collaboration/study-groups	course_id?: number	
	/collaboration/study-groups	{ name, course_id, description, max_members, meeting_schedule }	ת
	/collaboration/study-groups/{id}/join		ת
	/collaboration/study-groups/{id}/messages		ת ת
	/collaboration/study-groups/{id}/messages	{ message, attachment_url?: string }	ת ת



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```
import Echo from 'laravel-echo';
import Pusher from 'pusher-js';

window.Pusher = Pusher;

const echo = new Echo({
  broadcaster: 'reverb',
  key: process.env.VITE_REVERB_APP_KEY || 'netzertech-reverb-key',
  wsHost: process.env.VITE_REVERB_HOST || '127.0.0.1',
  wsPort: process.env.VITE_REVERB_PORT || 8080,
  wssPort: process.env.VITE_REVERB_PORT || 8080,
  forceTLS: false,
  enabledTransports: ['ws', 'wss'],
  authEndpoint: 'http://127.0.0.1:8080/api/v1/broadcasting/auth',
  auth: {
    headers: { Authorization: `Bearer ${localStorage.getItem('auth_token')}` }
  }
});

// Example: Listen for Live Study Group Messages
echo.private(`study-group.${groupId}`)
  .listen('.message.posted', (event) => {
    appendMessageToThread(event.message);
  });
```



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	ת	errors ת	ת
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